

Integer and Place-Value Fluency

Drill Book · Gauss Contest Preparation · used from Module 2 all term

Goal: Answer every item correctly in under 30 seconds.

How to use. One set per sitting. Start a timer, work straight through, stop the timer, then mark with the tear-off key at the back. Write your score and time in the box. Record the code of your most frequent mistake in the progress tracker. Repeat a set until you are at 90% or better within the target time. Every item should take under 45 seconds; anything longer belongs in homework, not here.

Reference page

M	HTh	TTh	Th	H	T	O
7	3	5	0	9	1	2

M millions · HTh hundred-thousands · TTh ten-thousands · Th thousands · H hundreds · T tens · O ones. Count places from the ones digit, in groups of three.

Area	Rule or fact
Place value	$10^1 = 10$, $10^2 = 100$, $10^3 = 1\,000$, $10^6 = 1\,000\,000$. $k \times 10^p$ is k followed by p zeros.
Comparing decimals	Pad to the same number of places, then compare: $0.35 = 0.350 > 0.305$.
Distance from 0	Ignore the sign: -8.2 is 8.2 from 0. Farthest from 0 is not the same as smallest.
Adding integers	$a + (-b) = a - b$. On a number line, + moves right, - moves left.
Subtracting integers	$a - (-b) = a + b$. The distance between a and b is (larger) - (smaller).
Crossing zero	From -11 to 6 is $11 + 6 = 17$: jump to 0, then jump on.
Time	$60\text{ s} = 1\text{ min}$, $60\text{ min} = 1\text{ h}$. Trade 60, never 100. Make the next hour first.
Money	$\$1 = 100\text{¢}$. Work in cents if the decimal point worries you.
Partner products	$2 \times 5 = 10$ · $4 \times 25 = 100$ · $8 \times 125 = 1\,000$ · $2 \times 50 = 100$ · $5 \times 20 = 100$.
Factor out	$a \times c + b \times c = (a + b) \times c$. Look for a shared number before multiplying.
Compensation	$998 = 1\,000 - 2$ and $1\,003 = 1\,000 + 3$: add the round parts, then the small parts.
Nines	$10^n - 1$ is n nines; its digit sum is 9n. $10^n - 47$ is (n - 2) nines then 53.
Die	Faces 1 to 6 add to 21. Opposite faces add to 7.
Digit sum	Add the digits; skip zeros; pair to 10 where you can.

Set 1 — single step, clean numbers

Target 6 min

Time _____

Score (of 20) _____

1. Which digit is in the tens place of 7 058 744?

2. Which is greater: 0.09 or 0.1?

3. Write 2×10^3 in standard form.

4. Which digit is in the tens place of 9 713 621?

5. Which is greater: 0.4 or 0.38?

6. Write 9×10^6 in standard form.

7. Which digit is in the hundred-thousands place of 2 012 236?

8. $-12 + 9 = ?$

9. $-5 - (-11) = ?$

10. How far apart are -10 and 3 on a number line?

11. $-11 + 9 = ?$

12. $3 - (-5) = ?$

13. How far apart are -6 and 6 on a number line?

14. $-7 + 6 = ?$

15. $125 \times 36 \times 8 = ?$

16. $1 \text{ h } 12 \text{ min} = \underline{\hspace{1cm}} \text{ min}$

17. What is the cost of 9 items at \$0.25 each?

18. $50 \times 30 \times 2 = ?$

19. $3 \text{ min } 53 \text{ s} = \underline{\hspace{1cm}} \text{ s}$

20. What is the cost of 7 items at \$0.75 each?

Set 2 — single step, clean numbers

Target 6 min

Time _____

Score (of 20) _____

1. Which digit is in the hundreds place of 1 295 754?

2. Which is greater: 0.6 or 0.599?

3. Write 2×10^4 in standard form.

4. Which digit is in the tens place of 3 277 951?

5. Which is greater: 0.5 or 0.49?

6. Write 4×10^4 in standard form.

7. Which digit is in the hundred-thousands place of 4 260 406?

8. $-14 + 9 = ?$

9. $-9 - (-14) = ?$

10. How far apart are -12 and 6 on a number line?

11. $-4 + 20 = ?$

12. $6 - (-8) = ?$

13. How far apart are -7 and 3 on a number line?

14. $-3 + 13 = ?$

15. $125 \times 44 \times 8 = ?$

16. $1 \text{ h } 55 \text{ min} = \underline{\hspace{1cm}} \text{ min}$

17. What is the cost of 12 items at \$0.50 each?

18. $125 \times 47 \times 8 = ?$

19. $4 \text{ min } 42 \text{ s} = \underline{\hspace{1cm}} \text{ s}$

20. What is the cost of 12 items at \$0.10 each?

Set 3 — two steps, contest-shaped numbers

Target 8 min

Time _____

Score (of 20) _____

1. What is the sum of the digits of 4 535 685?

2. Which is the middle value of 0.04, 0.4, 0.405?

3. $4 \times 10^4 + 7 \times 10^2 + 1 \times 10 = ?$

4. What is the sum of the digits of 2 760 553?

5. Which is the middle value of 0.3, 0.35, 0.03?

6. $5 \times 10^4 + 6 \times 10^2 + 6 \times 10 = ?$

7. What is the sum of the digits of 6 499 547?

8. $3 - (-9) + (-8) = ?$

9. It was -2°C . It rose 14 degrees, then fell 2 degrees. What is the temperature now?

10. Which is farthest from 0: 4.75, -4.95 , 5.2, -5.1 ?

11. $9 - (-5) + (-8) = ?$

12. It was -7°C . It rose 21 degrees, then fell 2 degrees. What is the temperature now?

13. Which is farthest from 0: -9.3 , 9.2 , -8.75 , 8.6 ?

14. $9 - (-6) + (-18) = ?$

15. $81 \times 30 + 19 \times 30 = ?$

16. What is the cost of 9 items at \$1.45 each?

17. $2 \text{ h } 49 \text{ min} + 3 \text{ h } 26 \text{ min} = ?$

18. A 180 cm strip is cut into 12 cm pieces; another 180 cm strip into 20 cm pieces.
How many more 12 cm pieces are there?

19. $17 \times 23 + 83 \times 23 = ?$

20. What is the cost of 12 items at \$1.75 each?

Set 4 — two steps, contest-shaped numbers

Target 8 min

Time _____

Score (of 20) _____

1. What is the sum of the digits of 6 245 658?

2. Which is the middle value of 0.65, 0.6, 0.06?

3. $7 \times 10^4 + 5 \times 10^2 + 4 \times 10 = ?$

4. What is the sum of the digits of 6 332 130?

5. Which is the middle value of 0.45, 0.04, 0.405?

6. $7 \times 10^4 + 3 \times 10^2 + 3 \times 10 = ?$

7. What is the sum of the digits of 9 420 717?

8. $10 - (-7) + (-14) = ?$

9. It was -10°C . It rose 24 degrees, then fell 9 degrees. What is the temperature now?

10. Which is farthest from 0: -5.75 , 5.8 , 6.1 , -6.15 ?

11. $8 - (-7) + (-16) = ?$

12. It was -5°C . It rose 17 degrees, then fell 6 degrees. What is the temperature now?

13. Which is farthest from 0: -7.4 , 6.85 , 7.2 , -6.7 ?

14. $14 - (-9) + (-13) = ?$

15. $44 \times 35 + 56 \times 35 = ?$

16. What is the cost of 9 items at $\$1.75$ each?

17. $3 \text{ h } 52 \text{ min} + 1 \text{ h } 39 \text{ min} = ?$

18. A 525 cm strip is cut into 15 cm pieces; another 525 cm strip into 25 cm pieces.
How many more 15 cm pieces are there?

19. $42 \times 37 + 58 \times 37 = ?$

20. What is the cost of 12 items at \$3.25 each?

Set 5 — reverse and multi-step

Target 10 min

Time _____

Score (of 20) _____

1. What number is 10 000 more than 3 047 000?

2. What is the sum of the digits of $10^5 - 25$?

3. $\underline{\hspace{1cm}} \times 10^4 = 620\,000$

4. What number is 10 000 more than 3 886 000?

5. What is the sum of the digits of $10^6 - 12$?

6. $\underline{\hspace{1cm}} \times 10^3 = 27\,000$

7. What number is 10 000 more than 1 262 000?

8. $\underline{\hspace{1cm}} - (-9) = 12$

9. After falling 11 degrees, the temperature was -4°C . What was it before the fall?

10. $-7 + \underline{\hspace{1cm}} = 2$

11. $\underline{\hspace{1cm}} - (-6) = 4$

12. After falling 11 degrees, the temperature was -2°C . What was it before the fall?

13. $-15 + \underline{\hspace{1cm}} = -5$

14. $\underline{\hspace{1cm}} - (-10) = 5$

15. A class ends at 3:08 p.m. after running 1 h 54 min. When did it start?

16. 4 pens cost \$8.60. What does one pen cost?

17. The five faces of a die that are not on the table add to 19. What number is on the bottom face?

18. A class ends at 11:02 a.m. after running 2 h 40 min. When did it start?

19. 6 pens cost \$8.70. What does one pen cost?

20. The five faces of a die that are not on the table add to 15. What number is on the bottom face?

Set 6 — reverse and multi-step

Target 10 min

Time _____

Score (of 20) _____

1. What number is 10 000 more than 2 352 000?

2. What is the sum of the digits of $10^5 - 3$?

3. $\underline{\hspace{1cm}} \times 10^2 = 7\,100$

4. What number is 10 000 more than 8 254 000?

5. What is the sum of the digits of $10^5 - 12$?

6. $\underline{\hspace{1cm}} \times 10^3 = 68\,000$

7. What number is 10 000 more than 3 355 000?

8. $\underline{\hspace{1cm}} - (-9) = 0$

9. After falling 15 degrees, the temperature was -7°C . What was it before the fall?

10. $-10 + \underline{\hspace{1cm}} = 12$

11. $\underline{\hspace{1cm}} - (-10) = 3$

12. After falling 17 degrees, the temperature was -2°C . What was it before the fall?

13. $-7 + \underline{\hspace{1cm}} = 10$

14. $\underline{\hspace{1cm}} - (-13) = 12$

15. A class ends at 3:05 p.m. after running 1 h 25 min. When did it start?

16. 5 pens cost \$8.75. What does one pen cost?

17. The five faces of a die that are not on the table add to 18. What number is on the bottom face?

18. A class ends at 11:05 a.m. after running 1 h 28 min. When did it start?

19. 7 pens cost \$15.05. What does one pen cost?

20. The five faces of a die that are not on the table add to 16. What number is on the bottom face?

Mixed final set — all three skill areas

Target 12 min

Time _____

Score (of 25) _____

1. What is the cost of 11 items at \$2.35 each?

2. What number is 10 000 more than 2 678 000?

3. A class ends at 12:10 p.m. after running 1 h 49 min. When did it start?

4. What is the cost of 8 items at \$0.25 each?

5. $-5 + \underline{\quad} = 7$

6. What is the sum of the digits of $10^7 - 12$?

7. Which is the middle value of 0.08, 0.805, 0.85?

8. Which is greater: 0.25 or 0.205?

9. $\underline{\hspace{1cm}} - (-5) = 2$

10. How far apart are -15 and 7 on a number line?

11. It was -6°C . It rose 23 degrees, then fell 6 degrees. What is the temperature now?

12. $\underline{\hspace{1cm}} \times 10^4 = 500\,000$

13. $3 - (-7) = ?$

14. Write 8×10^5 in standard form.

15. $15 - (-9) + (-14) = ?$

16. $5 \times 10^4 + 8 \times 10^2 + 4 \times 10 = ?$

17. 2 min 13 s = ____ s

18. After falling 19 degrees, the temperature was -3°C . What was it before the fall?

19. $-3 + 3 = ?$

20. 6 pens cost \$12.90. What does one pen cost?

21. $82 \times 23 + 18 \times 23 = ?$

22. $2 \text{ h } 47 \text{ min} + 3 \text{ h } 38 \text{ min} = ?$

23. Which digit is in the hundred-thousands place of 7 275 663?

24. $20 \times 42 \times 5 = ?$

25. Which is farthest from 0: -6.1 , 5.95 , 6.3 , -5.85 ?

Answer key — tear off along the binding before starting

#	Set 1	Set 2	Set 3	Set 4	Set 5	Set 6	Mixed
1	4	7	36	36	3 057 000	2 362 000	\$25.85
2	0.1	0.6	0.4	0.6	39	43	2 688 000
3	2 000	20 000	40 710	70 540	62	71	10:21 a.m.
4	2	5	28	18	3 896 000	8 264 000	\$2.00
5	0.4	0.5	0.3	0.405	52	43	12
6	9 000 000	40 000	50 660	70 330	27	68	61
7	0	2	44	30	1 272 000	3 365 000	0.805
8	−3	−5	4	3	3	−9	0.25
9	6	5	10°C	5°C	7°C	8°C	−3
10	13	18	5.2	−6.15	9	22	22
11	−2	16	6	−1	−2	−7	11°C
12	8	14	12°C	6°C	9°C	15°C	50
13	12	10	−9.3	−7.4	10	17	10
14	−1	10	−3	10	−5	−1	800 000
15	36 000	44 000	3 000	3 500	1:14 p.m.	1:40 p.m.	10
16	72	115	\$13.05	\$15.75	\$2.15	\$1.75	50 840
17	\$2.25	\$6.00	6 h 15 min	5 h 31 min	2	3	133
18	3 000	47 000	6	14	8:22 a.m.	9:37 a.m.	16°C
19	233	282	2 300	3 700	\$1.45	\$2.15	0
20	\$5.25	\$1.20	\$21.00	\$39.00	6	5	\$2.15
21							2 300
22							6 h 25 min
23							2
24							4 200
25							6.3